Architectural Dynamics of Event Density

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Abstract
Event Density (ED) is a minimal architectural framework that generates coherent structure from a simple dynamical rule. Previous work established the existence of ED's architectural invariants—saddles, channels, horizons, and cores—and formalized the ontology and regime taxonomy that govern ED dynamics. This paper introduces the physics layer of Event Density by defining and analyzing the energy-like and entropy-like quantities that arise naturally from the ED ontology. We define total mass, gradient energy, mobility energy, entropy, horizon flux, and core concentration, and we measure their behavior across the hyper-coherent, stable-coherent, marginal, and non-individuating regimes. These quantities reveal that ED supports a coherent physics: energy concentrates in cores, entropy drops during individuation, flux collapses at horizons, and mobility freezes as structures stabilize. The resulting behaviors mirror thermodynamic, statistical, and cosmological systems, demonstrating that ED's architectural motifs are also physical ones. This work establishes ED as a unified architectural-physical framework and provides the foundation for future developments in thermodynamics, statistical mechanics, and cosmology within the ED program.

Introduction
Event Density (ED) was introduced as an architectural framework: a minimal dynamical rule that generates structure, individuates coherent entities, and produces the characteristic motifs of saddles, channels, horizons, and cores. In the first paper, ED-Arch-01, the focus was on demonstrating that this architecture arises robustly and generically. In ED-Arch-02, the ontology and mathematics of the architecture were formalized, establishing the invariants, regime taxonomy, and geometric logic that govern ED dynamics. Together, these works showed that ED is not a model of a particular physical system but a universal architectural engine capable of producing coherent structure from minimal assumptions.

What ED does not yet possess is a physics. The architectural layer describes how structure forms, stabilizes, and interacts, but it does not specify what physical quantities live on that structure, how they evolve, or how they constrain the dynamics. Without such quantities, ED cannot be compared to thermodynamic systems, statistical mechanical ensembles, or cosmological processes. It cannot express conservation, dissipation, flux, or entropy. It cannot yet be interpreted as a physical theory.

This paper introduces the physics layer of Event Density by defining and analyzing a set of energy-like and entropy-like quantities that arise naturally from the ED ontology. These quantities—total mass, gradient energy, mobility energy, entropy, horizon flux, and core concentration—provide the first physical interpretation of ED dynamics. They reveal how ED stores, transports, and redistributes "energy" across its architectural structures, and how these processes differ across the hyper-coherent, stable-coherent, marginal, and non-individuating regimes.

The central idea is that ED's architectural motifs are not merely geometric features but physical ones. Saddles correspond to regions of high gradient energy; channels act as conduits for energy flow; horizons behave as semi-permeable boundaries that regulate flux; cores accumulate and concentrate energy in a manner analogous to condensed phases or gravitational wells. Mobility, previously a dynamical parameter, becomes interpretable as a kinetic-like quantity that governs the capacity for motion and deformation. Noise, previously a source of

stochasticity, becomes interpretable as an effective temperature.

By defining these quantities and measuring their behavior across simulations, ED-Phys-01 establishes the first energy accounting framework for Event Density. This framework enables three major advances. First, it provides a physical interpretation of ED dynamics, showing how architectural transitions correspond to changes in energy distribution, entropy, and flux. Second, it connects ED to thermodynamics and statistical mechanics, revealing analogues of phase transitions, condensation, freezing, and critical surfaces. Third, it creates a bridge to cosmology by showing how ED cores, channels, and horizons behave like physical structures with measurable energetic properties.

The goal of this paper is not to propose a specific physical model but to demonstrate that ED supports a coherent physics layer grounded in its architecture. The quantities defined here are not imposed externally; they arise naturally from the ED ontology and reveal the physical logic implicit in the architecture. With this layer in place, ED becomes a candidate framework for describing physical processes across scales, from condensed-matter-like behavior to cosmological structure formation.

The remainder of the paper defines the energy and entropy measures, describes how they are computed, analyzes their behavior across regimes, and interprets their physical significance. Together, these results show that ED is not only an architectural engine but a physical one—capable of supporting conserved quantities, fluxes, and thermodynamic behavior. This establishes the foundation for the next stage of the program: ED as a cosmological theory.

1. Physical Quantities in Event Density

The architectural layer of Event Density defines the geometry and dynamics of the system: gradients, saddles, channels, horizons, and cores. The physics layer introduces quantities that measure how the system stores, transports, and redistributes energy-like and entropy-like structure across this architecture. These quantities arise directly from the ED ontology and require no additional assumptions. Each one corresponds to a distinct physical intuition—mass, tension, kinetic capacity, disorder, flux, and concentration—and each one behaves differently across the ED regime taxonomy.

The quantities defined in this section form the foundation of ED's physical interpretation. They are computed at each timestep of the simulation and analyzed across regimes, transitions, and architectural motifs.

1.1 Total Mass

Total mass measures the amount of event density present in the system. It is the simplest physical quantity in ED and the only one that is strictly conserved in the absence of noise.

$$M = \sum_i \rho_i$$

Interpretation

Mass represents the total "participation" of the system. It is distributed unevenly across the lattice, concentrating in cores, thinning in voids, and flowing along channels. Mass conservation ensures that ED dynamics redistribute density but do not create or destroy it, except through noise-driven fluctuations.

Architectural role

Mass distribution reveals the large-scale structure of the system: the emergence of cores, the formation of channels, and the depletion of voids. It is the baseline against which all other energy-like quantities are interpreted.

1.2 Gradient Energy

Gradient energy measures the tension stored in spatial variations of the event density field. It is the first genuinely "energetic" quantity in ED.

$$E_\nabla = \sum_i |\nabla \rho_i|^2$$

Interpretation

Gradient energy quantifies how sharply the density field changes. High values indicate steep boundaries, saddle structures, and regions of architectural tension. Low values indicate smooth, homogeneous regions or the interior of stable cores.

Architectural role

Saddles are the geometric loci of gradient energy. Channels form along extended ridges of high gradient magnitude. Horizons emerge where gradient energy becomes sharply asymmetric across a boundary. Gradient energy therefore tracks the formation and evolution of ED's architectural motifs.

1.3 Mobility Energy

Mobility is a primitive in ED's dynamical rule. Mobility energy interprets this primitive physically by measuring the system's capacity for motion and deformation.

$$E_\mu = \sum_i \mu_i \rho_i$$

Interpretation

Mobility energy behaves like a kinetic-like quantity: it measures how much of the system is capable of moving, flowing, or reorganizing. High mobility energy corresponds to hyper-coherent regimes where structure is dynamic and responsive. Low mobility energy corresponds to stable-coherent regimes where structure freezes into persistent cores.

Architectural role

Mobility energy collapses inside mature cores and remains high along channels. It is a sensitive indicator of regime transitions, especially the onset of freezing and unfreezing.

1.4 Entropy-Like Measure

Entropy in ED measures the distributional disorder of the density field. It is defined using the normalized density $p_i = \rho_i / M$.

$$S = -\sum_i p_i \log p_i$$

Interpretation

Entropy is high when density is spread evenly across the lattice and low when density is concentrated in a few locations. Entropy therefore tracks the system's degree of order, coherence, and individuation.

Architectural role
Entropy drops sharply during core formation and rises during noise-driven diffusion. It distinguishes non-individuating regimes (high entropy) from coherent regimes (low entropy) and reveals the thermodynamic character of regime transitions.

1.5 Horizon Flux
Horizon flux measures the flow of density across architectural horizons. A horizon is a boundary where mobility collapses asymmetrically, creating a semi-permeable membrane. Flux across this boundary is defined using the ED flow vector J_i and the outward normal $\hat{n}_i$.

$$\Phi = \sum_{i \in H} J_i \cdot \hat{n}_i$$

Interpretation
Flux quantifies how much "stuff" crosses a horizon per unit time. High flux indicates active exchange between interior and exterior regions. Low flux indicates isolation or freezing.

Architectural role
Horizon flux is the key diagnostic of architectural individuation. In stable-coherent regimes, flux collapses and horizons become impermeable. In hyper-coherent regimes, flux remains high and horizons are dynamic. Flux therefore measures the permeability and dynamical status of ED's architectural boundaries.

1.6 Core Concentration Energy
Core concentration energy measures how strongly density accumulates inside cores. It is defined as a quadratic measure over the core region C.

$$E_{\text{core}} = \sum_{i \in C} \rho_i^2$$

Interpretation
This quantity captures the depth and stability of cores. Because it weights high densities more strongly, it grows rapidly as cores condense and stabilizes once cores reach equilibrium.

Architectural role
Core concentration energy tracks the life cycle of cores: formation, deepening, merging, and evaporation. It is the ED analogue of potential well depth in gravitational or condensed-matter systems.

1.7 Summary of Physical Roles
Each quantity corresponds to a distinct physical intuition and architectural function:

- Mass — total participation; conserved baseline.
- Gradient energy — architectural tension; saddles and boundaries.
- Mobility energy — kinetic-like capacity; freezing and unfreezing.

- Entropy — distributional disorder; individuation vs. diffusion.
- Horizon flux — permeability of boundaries; dynamical individuation.
- Core concentration — depth and stability of coherent entities.

Together, these quantities form the physics layer of ED. They allow ED to be interpreted thermodynamically, statistically, and cosmologically, and they provide the empirical foundation for the regime-by-regime analysis that follows.

2. Physics Across the ED Regimes

The regime taxonomy introduced in ED-Arch-02 describes how Event Density organizes itself architecturally: whether it individuates, stabilizes, diffuses, or oscillates. The physical quantities defined in Section 3 now allow these regimes to be understood as distinct physical phases, each characterized by specific patterns of energy, entropy, mobility, and flux.

Each regime expresses a coherent physical signature. These signatures are not imposed externally; they arise from the intrinsic dynamics of ED. The result is a phase-like structure analogous to thermodynamic, statistical, and cosmological systems, but grounded in ED's architectural ontology.

2.1 Hyper-Coherent Regime

The hyper-coherent regime is the phase of maximal dynamical activity. Structure forms rapidly, reorganizes continuously, and interacts strongly across the lattice.

Physical Characteristics

- Gradient energy is high due to steep boundaries and rapidly shifting saddles.
- Mobility energy is high; the system is fully capable of motion and deformation.
- Entropy is moderate: structure exists, but it is dynamic and not yet deeply individuated.
- Horizon flux is high; boundaries are permeable and constantly reconfigured.
- Core concentration energy is low to moderate; cores form but remain shallow and unstable.

Architectural Interpretation

Hyper-coherence corresponds to a pre-individuated phase where the system is actively searching the space of possible structures. Saddles are abundant, channels are fluid, and horizons are transient. The system behaves like a high-temperature fluid or a turbulent field.

Physical Analogy

- High-temperature plasma
- Turbulent fluid dynamics
- Early-universe pre-structure phase

2.2 Stable-Coherent Regime

The stable-coherent regime is the phase of deep individuation. Cores form, horizons stabilize, and the system develops persistent structure.

Physical Characteristics

- Gradient energy drops inside cores but remains high at boundaries.
- Mobility energy collapses in core interiors, indicating freezing.
- Entropy decreases sharply as density concentrates.
- Horizon flux collapses; horizons become semi-permeable or impermeable.
- Core concentration energy rises rapidly and stabilizes at high values.

Architectural Interpretation
Stable-coherence corresponds to the emergence of **persistent entities**. Cores behave like condensed phases or gravitational wells. Horizons act as protective boundaries that regulate exchange with the environment. Channels become stable conduits connecting cores.

Physical Analogy

- Condensed matter phases
- Gravitationally bound structures
- Low-temperature ordered systems

2.3 Marginal Regime
The marginal regime sits at the boundary between coherence and diffusion. It is characterized by oscillatory behavior, intermittent individuation, and sensitivity to noise.

Physical Characteristics

- Gradient energy oscillates as structures form and dissolve.
- Mobility energy fluctuates, reflecting partial freezing and unfreezing.
- Entropy hovers at intermediate values.
- Horizon flux is intermittent; boundaries appear and vanish.
- Core concentration energy rises and falls without stabilizing.

Architectural Interpretation
The marginal regime is a critical surface in ED's phase space. The system is neither fully individuated nor fully diffuse. Small perturbations can push it toward stable coherence or toward non-individuation.

Physical Analogy

- Critical points in phase transitions
- Edge-of-chaos systems
- Metastable states in statistical mechanics

2.4 Non-Individuating Regime
The non-individuating regime is the phase of diffusion and homogeneity. Structure fails to form or quickly dissolves.

Physical Characteristics

- Gradient energy is low; the field is smooth and lacks boundaries.

- Mobility energy is high but unstructured; motion does not produce coherence.
- Entropy is maximal; density is evenly distributed.
- Horizon flux is undefined or trivial because horizons do not form.
- Core concentration energy remains near zero.

Architectural Interpretation
Non-individuation corresponds to a high-entropy, high-temperature phase where the system lacks the architectural tension needed to form saddles, channels, or cores. Noise dominates, and the system behaves like a uniform field.

Physical Analogy

- High-temperature gas
- Thermalized equilibrium
- Cosmological radiation-dominated era

2.5 Regime Transitions as Physical Phase Transitions
The transitions between ED regimes exhibit the hallmarks of physical phase transitions:

- Hyper-coherent → Stable-coherent
    - Mobility collapse
    - Entropy drop
    - Core concentration spike
    - Horizon flux collapse
    - Analogue: condensation or gravitational collapse
- Stable-coherent → Hyper-coherent
    - Mobility resurgence
    - Gradient energy spike
    - Horizon dissolution
    - Analogue: melting or reheating
- Marginal ↔ Coherent
    - Oscillatory behavior
    - Sensitivity to noise
    - Analogue: critical phenomena
- Marginal ↔ Non-Individuating
    - Entropy plateau
    - Gradient energy collapse
    - Analogue: loss of order at high temperature
    -

These transitions reveal that ED's architectural regimes are not merely geometric categories but physical phases with distinct energetic signatures.

2.6 Summary of Regime Physics
Each regime expresses a coherent physical identity:

- Hyper-coherent — high energy, high flux, dynamic structure.
- Stable-coherent — low entropy, deep cores, frozen horizons.
- Marginal — oscillatory, critical, metastable.

- Non-individuating — diffuse, high entropy, structureless.

These signatures demonstrate that ED supports a rich physics layer grounded in its architecture. The next sections will quantify these behaviors empirically and interpret their implications for thermodynamics, statistical mechanics, and cosmology.

3. Physical Interpretation

The physical quantities defined in Section 3 reveal that Event Density is not merely a geometric or architectural system. It expresses behaviors that are recognizably physical: conservation, flux, tension, condensation, freezing, diffusion, and the emergence of boundaries that regulate exchange. These behaviors arise intrinsically from the ED ontology and do not require external physical assumptions. This section interprets ED through three complementary lenses—thermodynamics, statistical mechanics, and cosmology—showing how the architecture supports a coherent physics layer.

3.1 Thermodynamic Interpretation

The thermodynamic interpretation begins with the observation that ED supports analogues of energy, entropy, temperature, and phase transitions.

Energy and Tension

Gradient energy behaves like a field-theoretic tension term. Regions of high gradient energy correspond to steep boundaries, saddle structures, and architectural stress. As the system evolves, gradient energy is redistributed, minimized in core interiors, and concentrated at horizons. This mirrors the thermodynamic tendency of systems to reduce free energy by smoothing gradients or forming stable structures.

Kinetic-like Capacity

Mobility energy plays the role of a kinetic-like term. High mobility corresponds to high effective temperature: the system is fluid, dynamic, and capable of reorganizing. Low mobility corresponds to low effective temperature: the system freezes into stable configurations. The collapse of mobility energy during core formation is analogous to a cooling-driven phase transition.

Entropy and Order

The entropy-like measure tracks the distributional disorder of the density field. High entropy corresponds to diffuse, homogeneous states; low entropy corresponds to concentrated, individuated states. The sharp entropy drop during core formation is the thermodynamic signature of ordering, condensation, or crystallization.

Phase Transitions

Regime transitions in ED exhibit the hallmarks of thermodynamic phase transitions:

- Hyper-coherent → Stable-coherent: condensation, freezing, ordering
- Stable-coherent → Hyper-coherent: melting, reheating, disordering
- Marginal regimes: critical surfaces with oscillatory, metastable behavior

These transitions are not imposed; they emerge from the interplay of gradient energy, mobility, and entropy.

3.2 Statistical Mechanical Interpretation

From a statistical mechanical perspective, ED behaves like a system of interacting degrees of freedom evolving

under a combination of deterministic and stochastic rules.

Noise as Effective Temperature
Noise in ED plays the role of an effective temperature. High noise levels prevent individuation, maintain high entropy, and keep the system in a non-individuating regime. Low noise levels allow structure to form, stabilize, and persist. Intermediate noise levels produce marginal, metastable behavior.

Cores as Condensed Phases
Cores behave like condensed phases or bound states. They accumulate density, minimize gradient energy, and suppress mobility. Their interiors are low-entropy, low-temperature regions separated from the environment by horizons that regulate exchange.

Horizons as Critical Surfaces
Horizon boundaries behave like critical surfaces or domain walls. They separate regions of different mobility, entropy, and concentration. Flux across horizons measures the permeability of these boundaries and determines whether cores grow, shrink, or remain stable.

Channels as Transport Pathways
Channels act as transport pathways for mass, energy, and flux. They are the ED analogue of conduction paths, filaments, or low-resistance routes in statistical systems. Their formation and dissolution reflect the system's attempt to optimize transport under architectural constraints.

Ensemble Behavior
Across many simulations, ED exhibits ensemble-level regularities: distributions of core sizes, flux patterns, and entropy trajectories. These regularities suggest that ED supports a statistical mechanical description with well-defined macroscopic observables emerging from microscopic rules.

3.3 Cosmological Interpretation
The cosmological interpretation emerges from the structural parallels between ED's architecture and the large-scale structure of the universe.

Cores as Gravitational Wells
Cores behave like gravitationally bound structures: they accumulate mass, deepen over time, and resist disruption. Their concentration energy grows as density collapses inward, analogous to gravitational potential energy becoming more negative as matter condenses.

Channels as Filaments
Channels resemble cosmological filaments: extended structures that connect cores, transport mass, and define the skeleton of large-scale structure. Their formation reflects the system's tendency to organize mass along coherent pathways.

Horizons as Causal Boundaries
Horizons behave like causal boundaries or event horizons. They regulate the flow of mass and energy, isolate interior regions, and define the effective "causal domain" of a core. Horizon flux measures the degree of coupling between interior and exterior regions.

Gradient Energy as Curvature

Gradient energy plays a role analogous to curvature or gravitational potential gradients. High gradient energy corresponds to steep density contrasts, which in cosmology correspond to regions of strong gravitational influence.

Regime Evolution as Cosmic Epochs
The regime taxonomy maps naturally onto cosmological epochs:

- Non-individuating → radiation-dominated era
- Hyper-coherent → matter-dominated structure formation
- Stable-coherent → halo formation and virialization
- Marginal → transitional epochs or environments near criticality

These analogies do not imply that ED is a cosmological model, but they show that ED's architecture supports cosmological-like behavior.

3.4 Synthesis: ED as a Physical Architecture

The thermodynamic, statistical, and cosmological interpretations converge on a single insight: ED's architecture is inherently physical. The same structures that define ED's geometry—saddles, channels, horizons, cores—also define its physics. Energy, entropy, flux, and concentration are not external additions but intrinsic consequences of the architectural invariants.

This synthesis establishes ED as a candidate framework for describing physical processes across scales. It provides a unified language in which architecture and physics are inseparable: geometry determines dynamics, dynamics determine energy flow, and energy flow determines structure.

4. Methods and Simulation Details

The physical interpretation of Event Density relies on quantitative measurements of mass, gradient energy, mobility energy, entropy, horizon flux, and core concentration. This section describes the computational methods used to obtain these quantities, the procedures for identifying architectural structures, and the simulation parameters used throughout the study. All results in this paper are reproducible from the definitions and methods provided here.

4.1 Simulation Framework

All simulations are performed on a discrete lattice with periodic boundary conditions. Each lattice site i stores two primary variables:

- **Event density** $\rho_i(t)$
- **Mobility** $\mu_i(t)$

The system evolves according to the ED update rule:

$$\rho_i(t+1) = \rho_i(t) + \Delta_i\,[\rho(t),\mu(t)] + \eta_i(t)$$

where Δ_i is the deterministic ED flow operator and $\eta_i(t)$ is stochastic noise.

Deterministic Flow

The deterministic component redistributes density according to local gradients and mobility:

$$\Delta_i = \sum_{j \in N(i)} \mu_j (\rho_j - \rho_i)$$

where $N(i)$ denotes the neighborhood of site i. This operator drives density from high-density regions toward low-density regions, modulated by mobility.

Noise
Noise is drawn from a zero-mean distribution:

$$\eta_i(t) \sim N(0,\sigma^2)$$

Noise amplitude σ acts as an effective temperature, controlling the system's ability to individuate or diffuse.

4.2 Gradient Computation
Gradient energy requires a spatial gradient operator. Gradients are computed using central differences:

$$\nabla \rho_i = \left(\frac{\rho_{i+\hat{x}} - \rho_{i-\hat{x}}}{2}, \frac{\rho_{i+\hat{y}} - \rho_{i-\hat{y}}}{2} \right)$$

The squared magnitude is:

$$|\nabla \rho_i|^2 = \left(\frac{\rho_{i+\hat{x}} - \rho_{i-\hat{x}}}{2} \right)^2 + \left(\frac{\rho_{i+\hat{y}} - \rho_{i-\hat{y}}}{2} \right)^2$$

This operator is used in the computation of gradient energy and in the detection of saddles and boundaries.

4.3 Flow Vector and Flux Measurement
The ED flow vector J_i is defined as:

$$J_i = \mu_i \nabla \rho_i$$

Flux across a horizon H is computed as:

$$\Phi = \sum_{i \in H} J_i \cdot \hat{n}_i$$

where $\hat{n}_i$ is the outward normal to the horizon boundary. Normals are computed using local gradient directions. Flux is measured at each timestep to track the permeability of horizons and the dynamical status of cores.

4.4 Core Detection
Cores are defined as connected regions of high density and low mobility. A site i is considered part of a core if:

$$\rho_i > \theta_\rho \quad \text{and} \quad \mu_i < \theta_\mu$$

where θ_ρ and θ_μ are thresholds determined empirically from the distribution of values in the simulation.

Connected components are identified using a flood-fill algorithm. Each component is treated as a core, and its properties (mass, concentration energy, flux boundary) are computed.

4.5 Horizon Detection

Horizons are boundaries where mobility collapses asymmetrically. A horizon site satisfies:

$$\mu_i < \theta_\mu \quad \text{and} \quad \exists j \in N(i) : \mu_j > \theta_\mu$$

This identifies the interface between a low-mobility interior and a higher-mobility exterior.

Horizon geometry is extracted by tracing the boundary of each core. Normals are computed using local gradient directions, enabling flux measurement.

4.6 Channel Detection

Channels are elongated structures of moderate density and high mobility. A site belongs to a channel if:

$$\rho_{min} < \rho_i < \rho_{max} \quad \text{and} \quad \mu_i > \theta_\mu$$

Channels are identified using ridge-detection techniques applied to the gradient magnitude field. Their connectivity and length are measured to characterize transport pathways.

4.7 Computation of Physical Quantities

The physical quantities defined in Section 3 are computed as follows:

Total Mass

$$M = \sum_i \rho_i$$

Gradient Energy

$$E_\nabla = \sum_i |\nabla \rho_i|^2$$

Mobility Energy

$$E_\mu = \sum_i \mu_i \rho_i$$

Entropy

$$S = -\sum_i p_i \log p_i, \quad p_i = \frac{\rho_i}{M}$$

Horizon Flux

$$\Phi = \sum_{i \in H} J_i \cdot \hat{n}_i$$

Core Concentration Energy

$$E_{\text{core}} = \sum_{i \in C} \rho_i^2$$

All quantities are computed at each timestep to produce time-series data for regime analysis.

4.8 Simulation Parameters

Unless otherwise noted, simulations use the following parameters:

- Lattice size: 128×128
- Neighborhood: 4-connected (von Neumann)
- Time steps: 10,000
- Noise amplitude: $\sigma \in [0.0, 0.2]$
- Mobility initialization: uniform or gradient-biased
- Density initialization: random or structured

Parameter sweeps are performed across noise amplitude, mobility initialization, and density initialization to explore the full regime taxonomy.

4.9 Reproducibility and Code Availability

All simulations can be reproduced from the definitions and methods provided here. The update rule, gradient operator, flux computation, and structure-detection algorithms are fully specified. Code implementing these methods is available in the accompanying repository.

5. Results

The physical quantities defined in Section 3 reveal distinct, measurable behaviors across the ED regime taxonomy. These behaviors are robust across initial conditions, noise levels, and mobility configurations. This section presents the empirical results obtained from simulations, focusing on the time evolution of energy-like and entropy-like quantities, the dynamics of core formation, the behavior of flux across horizons, and the signatures of regime transitions.

The results demonstrate that ED supports a coherent physics layer: energy concentrates, entropy drops, flux collapses, and mobility freezes in ways that mirror thermodynamic, statistical, and cosmological systems.

5.1 Time Evolution of Physical Quantities

Across all simulations, the time evolution of the physical quantities exhibits consistent patterns that correspond to the four ED regimes.

Gradient Energy

Gradient energy rises sharply during early structure formation, peaks during saddle proliferation, and then either stabilizes (in stable-coherent regimes) or decays (in non-individuating regimes). In hyper-coherent regimes, gradient energy remains elevated due to persistent architectural tension.

Mobility Energy

Mobility energy is initially high in all simulations. In stable-coherent regimes, it collapses rapidly as cores form

and freeze. In hyper-coherent regimes, mobility energy remains high, reflecting continuous reorganization. In marginal regimes, mobility energy oscillates, tracking intermittent freezing and unfreezing.

Entropy
Entropy decreases sharply during core formation, stabilizing at low values in stable-coherent regimes. In non-individuating regimes, entropy remains high throughout. In marginal regimes, entropy fluctuates around intermediate values, reflecting partial individuation.

Core Concentration Energy
Core concentration energy grows rapidly once cores nucleate. In stable-coherent regimes, it reaches a plateau corresponding to mature, persistent cores. In hyper-coherent regimes, it rises and falls as cores form and dissolve. In non-individuating regimes, it remains near zero.

Horizon Flux
Flux across horizons collapses sharply during core stabilization. In hyper-coherent regimes, flux remains high due to permeable, dynamic boundaries. In marginal regimes, flux oscillates, reflecting intermittent horizon formation. These time-series behaviors provide quantitative signatures for each regime.

5.2 Core Formation and Evolution
Core formation is a central architectural process in ED, and the physical quantities reveal its dynamics with precision.

Nucleation
Cores nucleate at locations of high gradient energy and moderate mobility. Early cores are shallow, with low concentration energy and high flux across their boundaries.

Growth
As cores accumulate density, concentration energy increases quadratically. Mobility collapses inside the core, and entropy drops locally. Horizons form around the core as mobility gradients sharpen.

Stabilization
In stable-coherent regimes, cores reach a steady state characterized by:

- low interior gradient energy
- low interior mobility
- high concentration energy
- near-zero horizon flux

These signatures indicate deep individuation.

Merging
When two cores approach, their horizons interact. If flux remains high, cores may merge, producing a spike in gradient energy followed by stabilization. If flux is low, cores remain isolated.

Evaporation
In hyper-coherent or marginal regimes, cores may dissolve. Concentration energy drops, mobility rises, and entropy increases. Evaporation is typically preceded by a resurgence of horizon flux.

These behaviors demonstrate that core dynamics are governed by the interplay of gradient energy, mobility, and flux.

5.3 Horizon Dynamics and Flux Behavior

Horizons are the architectural boundaries that regulate exchange between interior and exterior regions. Their behavior is one of the clearest physical signatures of ED regimes.

Flux Collapse

In stable-coherent regimes, flux across horizons collapses to near zero. This indicates that cores become isolated, with minimal exchange of density or mobility. Flux collapse is a reliable indicator of individuation.

Flux Persistence

In hyper-coherent regimes, flux remains high. Horizons are permeable, and cores do not fully isolate. This corresponds to a high-temperature, high-mobility phase.

Flux Oscillation

In marginal regimes, flux oscillates. Horizons form and dissolve repeatedly, reflecting metastable individuation.

Flux Absence

In non-individuating regimes, horizons do not form, and flux is undefined or trivial.

Flux behavior therefore provides a direct measure of architectural individuation and boundary stability.

5.4 Regime Signatures in Physical Quantities

Each regime exhibits a distinct pattern across the physical quantities:

Hyper-Coherent

- High gradient energy
- High mobility energy
- Moderate entropy
- High horizon flux
- Low core concentration

Stable-Coherent

- Low interior gradient energy, high boundary gradient energy
- Low mobility energy inside cores
- Low entropy
- Near-zero horizon flux
- High core concentration

Marginal

- Oscillatory gradient energy
- Oscillatory mobility energy
- Intermediate entropy
- Intermittent horizon flux
- Fluctuating core concentration

Non-Individuating

- Low gradient energy
- High mobility energy
- High entropy
- No horizons
- No core concentration

These signatures are robust across parameter sweeps and initial conditions.

5.5 Phase-Transition-Like Behavior

Regime transitions exhibit sharp changes in physical quantities that resemble phase transitions:

Hyper-Coherent → Stable-Coherent

- Mobility collapse
- Entropy drop
- Flux collapse
- Concentration spike

Stable-Coherent → Hyper-Coherent

- Mobility resurgence
- Gradient energy spike
- Horizon dissolution

Marginal ↔ Coherent

- Oscillatory behavior
- Sensitivity to noise

Marginal ↔ Non-Individuating

- Entropy plateau
- Gradient energy collapse

These transitions demonstrate that ED's architectural regimes correspond to physical phases with distinct energetic and entropic signatures.

5.6 Ensemble-Level Regularities

Across many simulations, ensemble-level patterns emerge:

- Core size distributions follow heavy-tailed patterns.
- Flux distributions cluster by regime.
- Entropy trajectories converge to regime-specific attractors.
- Gradient energy exhibits universal early-time growth followed by regime-dependent stabilization.

These regularities suggest that ED supports a statistical mechanical description with well-defined macroscopic observables.

5.7 Summary of Results

The results demonstrate that ED's architectural structures—saddles, channels, horizons, and cores—are accompanied by distinct physical signatures. Energy concentrates, entropy drops, flux collapses, and mobility freezes in ways that mirror thermodynamic, statistical, and cosmological systems. These behaviors confirm that ED supports a coherent physics layer grounded in its architecture.

6. Discussion and Implications

The results presented in this paper demonstrate that Event Density supports a coherent physics layer grounded in its architectural ontology. The physical quantities defined in Section 3—mass, gradient energy, mobility energy, entropy, horizon flux, and core concentration—exhibit robust, interpretable behaviors across the ED regime taxonomy. These behaviors reveal that ED is not merely a geometric engine but a physical architecture capable of supporting conserved quantities, fluxes, phase transitions, and emergent structures analogous to those found in thermodynamics, statistical mechanics, and cosmology.

This section discusses the implications of these findings, focusing on three themes: the emergence of physical behavior from architectural invariants, the unification of geometry and physics within ED, and the potential of ED as a framework for understanding physical systems across scales.

6.1 Architecture as the Source of Physics

A central insight of this work is that ED's physical behavior arises directly from its architectural invariants. The same structures that define ED's geometry—saddles, channels, horizons, and cores—also define its physics.

- Gradient energy concentrates at saddles and boundaries, revealing architectural tension.
- Mobility energy collapses inside cores, revealing architectural freezing.
- Entropy drops during individuation, revealing architectural ordering.
- Flux collapses at horizons, revealing architectural isolation.
- Concentration energy grows in cores, revealing architectural condensation.

These correspondences show that ED's architecture is not a passive geometric scaffold but an active generator of physical behavior. The architectural layer and the physics layer are inseparable: geometry determines dynamics, dynamics determine energy flow, and energy flow determines structure.

This architectural determinism is a distinctive feature of ED. Unlike traditional physical theories, where geometry and dynamics are specified separately, ED derives both from a single underlying rule. This unification suggests that ED may provide a new way of thinking about physical systems in which architecture and physics are fundamentally intertwined.

6.2 ED as a Unified Physical Framework

The physical behaviors observed in ED span multiple domains of physics:

- Thermodynamics: entropy reduction, energy minimization, phase transitions
- Statistical mechanics: noise as temperature, metastability, ensemble regularities
- Cosmology: core condensation, filamentary channels, horizon-like boundaries

The fact that a single architectural rule produces analogues of these behaviors suggests that ED may serve as a unifying framework for understanding physical systems across scales. ED does not reproduce the detailed equations of thermodynamics, statistical mechanics, or cosmology, but it reproduces their structural logic: the way

systems organize, stabilize, and exchange energy.

This structural universality is significant. It implies that ED captures something fundamental about how physical systems generate and maintain structure. The architectural motifs of ED—saddles, channels, horizons, and cores—may represent universal organizational patterns that appear across many physical domains.

6.3 Implications for Thermodynamics and Statistical Mechanics

The thermodynamic and statistical mechanical interpretations of ED suggest several implications:

- Entropy as architectural disorder: ED provides a geometric interpretation of entropy as the distributional disorder of event density.
- Temperature as mobility/noise: ED links effective temperature to mobility and noise, providing a dynamical interpretation of thermal behavior.
- Phase transitions as regime transitions: ED's regime transitions exhibit the hallmarks of first-order and continuous phase transitions.
- Cores as condensed phases: ED cores behave like condensed phases or bound states, with low entropy and low mobility.
- Horizons as domain walls: ED horizons act as semi-permeable boundaries that regulate flux, analogous to domain walls in statistical systems.

These correspondences suggest that ED may offer a new perspective on thermodynamic and statistical phenomena, one grounded in architectural invariants rather than microscopic particle interactions.

6.4 Implications for Cosmology

The cosmological interpretation of ED is particularly striking. Without any explicit gravitational dynamics, ED produces structures that resemble the large-scale structure of the universe:

- Cores behave like gravitational wells or halos.
- Channels resemble cosmic filaments.
- Horizons behave like causal boundaries.
- Gradient energy plays a role analogous to curvature or gravitational potential gradients.
- Regime evolution mirrors cosmological epochs.

These analogies do not imply that ED is a cosmological model, but they suggest that ED captures the architectural logic of structure formation. This raises the possibility that cosmological structure may arise from architectural principles more general than gravity itself.

6.5 Toward a General Theory of Physical Architecture

The results of this paper point toward a broader conceptual implication: physical systems may be governed by architectural principles that transcend specific forces or equations. ED provides a concrete example of such a principle. From a minimal rule, ED generates:

- coherent entities
- stable boundaries
- transport pathways
- energy concentration
- entropy reduction

- phase transitions
- hierarchical structure

These features are hallmarks of physical systems across scales. ED suggests that these features may arise from universal architectural invariants rather than domain-specific mechanisms.

This perspective opens the door to a new kind of physical theory—one in which architecture is primary and dynamics emerge from architectural constraints.

6.6 Limitations and Future Directions
While ED exhibits rich physical behavior, several limitations remain:

- ED does not yet incorporate explicit forces or interactions beyond mobility-weighted flow.
- The relationship between ED's architectural invariants and known physical laws remains to be formalized.
- The mapping between ED regimes and physical phases is qualitative rather than quantitative.
- The cosmological analogies, while compelling, require further development and mathematical grounding.

These limitations point toward future work:

- Introducing explicit force-like terms or coupling rules
- Formalizing the thermodynamic interpretation
- Deriving ensemble-level predictions
- Developing ED as a cosmological framework

The present paper establishes the foundation for these developments.

6.7 Synthesis
The physical interpretation of ED reveals that architecture and physics are deeply intertwined. ED's architectural invariants generate physical behavior that spans thermodynamics, statistical mechanics, and cosmology. The physical quantities defined in this paper provide a coherent framework for understanding these behaviors, and the results demonstrate that ED supports a rich physics layer grounded in its architecture.

This synthesis positions ED as a candidate framework for understanding physical systems across scales, offering a unified perspective in which geometry, dynamics, and energy flow emerge from a single architectural rule.

7. Conclusion
Event Density was introduced as an architectural framework capable of generating coherent structure from a minimal dynamical rule. The first two papers established the existence of this architecture and formalized its ontology, mathematics, and regime taxonomy. The present work extends ED from architecture to physics by defining and analyzing the energy-like and entropy-like quantities that arise naturally from the ED ontology.

The physical quantities introduced in this paper—mass, gradient energy, mobility energy, entropy, horizon flux, and core concentration—reveal that ED supports a coherent physics layer grounded in its architectural invariants. These quantities exhibit robust, interpretable behaviors across the ED regime taxonomy. They concentrate, collapse, oscillate, and stabilize in ways that mirror thermodynamic, statistical, and cosmological systems. They

provide measurable signatures of individuation, boundary formation, transport, and condensation.

The results demonstrate that ED's architectural motifs—saddles, channels, horizons, and cores—are not merely geometric features but physical ones. Saddles store gradient energy; channels transport mass and mobility; horizons regulate flux; cores accumulate and concentrate energy. These correspondences show that ED's architecture and physics are inseparable: geometry determines dynamics, dynamics determine energy flow, and energy flow determines structure.

This synthesis positions ED as a candidate framework for understanding physical systems across scales. ED does not reproduce the detailed equations of thermodynamics, statistical mechanics, or cosmology, but it reproduces their structural logic. It exhibits phase-transition-like behavior, supports condensed-phase-like cores, forms filamentary channels, and generates horizon-like boundaries. These behaviors arise from a single architectural rule, suggesting that physical organization may be governed by architectural principles more fundamental than domain-specific forces.

By establishing ED's physics layer, this paper provides the foundation for the next stage of the ED program. Future work will explore explicit force-like extensions, ensemble-level predictions, thermodynamic formalisms, and cosmological interpretations. The architectural invariants identified in ED-Arch-02 and the physical quantities defined here will serve as the basis for these developments.

Event Density now stands as a unified architectural-physical framework: a system in which structure, dynamics, and energy flow emerge from a single generative principle. The physics introduced in this paper transforms ED from a structural engine into a physical theory, opening the path toward a general science of physical architecture.

8. Future Work

The physics layer introduced in this paper establishes the foundation for a broader scientific program in Event Density. Several directions follow naturally from the results presented here.

Extensions of the Physics Layer

The physical quantities defined in this work—gradient energy, mobility energy, entropy, flux, and concentration—suggest additional quantities that may further illuminate ED's dynamics. These include curvature-like measures derived from second-order gradients, stress-like tensors associated with anisotropic flows, and potential-like quantities associated with long-range architectural interactions. Formalizing these extensions would deepen the connection between ED and field-theoretic descriptions.

Thermodynamic and Statistical Formulations

The entropy and mobility behaviors observed across regimes indicate that ED supports a thermodynamic interpretation. Future work will formalize this interpretation by deriving effective equations of state, identifying thermodynamic potentials, and characterizing phase transitions quantitatively. Ensemble-level analyses may reveal universal scaling laws, critical exponents, or fluctuation–dissipation relations.

Force-Like and Interaction-Like Extensions

The present formulation of ED relies solely on mobility-weighted flow. Introducing explicit force-like or interaction-like terms—while preserving architectural invariants—may allow ED to model a broader class of physical systems. Such extensions could include attraction–repulsion rules, curvature-dependent mobility, or coupling between density and mobility fields.

Cosmological Development

The architectural parallels between ED and cosmological structure formation suggest a path toward an ED-based cosmology. Future work will explore the mapping between ED cores and halos, channels and filaments, horizons and causal boundaries, and gradient energy and curvature. This may lead to an ED-Cosmo framework capable of reproducing large-scale structure from architectural principles.

Analytical Theory

While this paper focuses on simulation-based results, the simplicity of the ED rule invites analytical treatment. Future work will develop continuum approximations, derive partial differential equations that approximate ED dynamics, and analyze fixed points, stability, and bifurcations. Analytical results would complement simulation findings and clarify the mathematical structure of ED's physics layer.

Multi-Scale and Multi-Layer Architectures

ED's architectural invariants may extend naturally to multi-scale or multi-layer systems. Future work will explore hierarchical ED models in which cores act as higher-level nodes, channels form higher-level networks, and horizons define nested boundaries. Such models may provide a unified architectural description of systems that exhibit structure across scales.

Together, these directions outline the next stage of the ED program: the development of a general theory of physical architecture grounded in the invariants, dynamics, and energetic behaviors revealed in this work.